



ISO/TC 12, Quantities and units

ISO/TC 12 N 1201

2017-11-14

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Minutes of the 14th meeting of ISO/TC 12 held on 2017-10-20 in Berlin, Germany

1. Opening of the meeting (09:00 h)

The ISO/TC12 Chairman, Prof. Pendrill (Sweden) welcomed the delegates and opened the meeting.

2. Roll call of delegates

A roll call of the P-, O- and L-Member attendance was taken. Delegations were in attendance from 6 P-members: Egypt, Finland, France, Germany, Sweden and USA; and one ISO/TC in liaison: ISO/TC 215, *Health informatics*. See also the attendance list circulated after the meeting in document N 1198.

In addition to the members present in the meeting room, Prof. Jacobson (DE), Dr. Ehrlich (US), Prof. Mari (IEC/TC 25 secretary) participated via Webex.

The secretariat support, Ms. Lindor Norén stressed that it is very important to register on the [ISO meetings](#) portal before the meeting.

3. Adoption of the agenda

The agenda, documents N 1182, was adopted as presented, however, agenda item 6 was moved to the afternoon and the sub items under item 9 were renumbered. The Chairman announced that due to cancelled flights he would have to leave the meeting at 15:00 h. ISO/TC 12 agreed that Prof. Grimvall would chair the meeting after 15:00 h.

Resolution 102, Berlin 1/2017

4. Appointment of the drafting committee

The Chairman asked for volunteers for the resolutions drafting committee. ISO/TC 12 thanked Ms. Lindor Norén (Sweden) and Ms. Saxholm (Finland) for volunteering and appointed them, along with Secretary Mr. Radhe (Sweden), to the resolutions drafting committee to draft the resolutions of the 14th meeting of ISO/TC 12.

Resolution 103, Berlin 2/2017



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5. Report of the Chair and of the Secretariat

The Chairman presented his report, issued as document N 1191. He noted that apart from producing its own standards, TC12 assists in reviewing use of quantities and units in other ISO standards and similar documents and encourages input from third parties to influence its own standards development.

He stated that quality assurance of metrological data in more qualitative situations and for decision-making, e.g. where humans (as customers, patients etc) are centre of focus, is still in its infancy. When the topic develops it would be a topic for future discussion ISO/TC 12.

The secretary presented document N 1180, summarizing updates to the membership, structure, and work programme of the committee. He noted that Poland is P-member in IEC/TC 25, *Quantities and units*, and O-member in ISO/TC 12. As the two committees have similar scope the secretary will approach the Polish member body and propose that they become P-members also in ISO/TC 12.

With respect to the secretary's report, ISO/TC 12 requested that the secretary should evaluate the participation of the member bodies of ISO/TC 12 and the participation of the WG 19 experts. For the final editing of the ISO 80000 series, the TPM and the secretary of WG 19 will also involve the respective project leaders.

Action 1, Berlin 2017: Secretary to assess the participation in TC 12 during the last 3 years. By April 1, 2018. This assessment shall also include WG 19.

Resolution 104, Berlin 3/2017

6. Report of the working group

6.1 Report of ISO/TC 12/WG 19, Revision ISO 80000, all parts

The convenor, Dr. Krystek, gave an oral report in the progress of WG 19. Mr. Krystek reported that WG 19 has had 2 meetings, in total of 5 days this year. It was noted that it would be preferable to hold WG meetings in conjunction to the TC 12 plenary meeting. ISO/TC 12 thanked him for his report and for the extensive work of the working group.

7. Business Plan for ISO/TC 12

The secretary presented the updated business plan for ISO/TC 12, doc. N 1178.

Action 2, Berlin 2017: ISO/TC 12 asks its TPM to see too that the current Business Plan is uploaded on iso.org.

Resolution 105, Berlin 4/2017



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8. Report on the work of ISO/TC 215 TF1, Task Force on Quantities and Units to be used in e-health Doc. ISO/TC 12 N 1190

The chairman presented doc N 1190. He pointed out that there was a task force in action for the coordinated effort between IEC/TC 25, ISO/TC 12 and IEC/TC 215 on quantities and units in e-health. The group has a very broad participation ranging from Australia, Japan, USA and Europe.

UCOM, which is referred to in the presentation, is a coordinated compilation of quantities and units including a codification to facilitate computerized handling. One aim is to have unique symbols for each quantity.

The final report will give a recommendation for additional standardization.

Mr. Krystek asked about the formation of the Task Force. ISO/TC 12 resolution 75, N 1100 and N 1111 address this issue. The result was that the combined vote between ISO/TC 12, ISO/TC 215 and IEC/TC 25 was in favour of the constitution of the Task Force.

9. Review of liaisons

9.1 ISO committees in liaison

JTC 1/SC 32, TC 8/SC 6, TC 37, TC 43, TC 43/SC 1, TC 43/SC 3, TC 47, TC 69, TC 85, TC 115, TC 203, TC 215

The Report from ISO/TC 69, *Application of statistical methods*, N 1170 was studied. At the June 2017 meeting in Cape Town, TC 69 resolved that WG 4 *Big Data Analytics* would be created. Initially three NWIP's are being created: *Big Data Terminology*, *CRISP DM Ver. 2* and *Model Validation*. This group has a close liaison relationship with ISO/JTC1/WG9, *Information Technology Big Data*.

If experts from ISO/TC 12 are interested in taking part in these projects, please contact the Secretary, sven.radhe@sis.se For further information, please contact the secretary of ISO/TC 69 at jadmussen@asq.org.

ISO/TC 69 is also involved in JCGM work. The chairman developed the issue of JCGM 100, *Evaluation of measurement data — Guide to the expression of uncertainty in measurement*.

Liaison reports were also received from ISO/TC 43, *Acoustics*, doc. N 1186, and ISO/TC 43/SC 1, *Noise*, doc. N 1187.

See §8 for liaison report from ISO/TC 215 Health informatics.

9.2 IEC committees in liaison

TC 1, TC 25

The secretary gave a brief oral report from the recent plenary meeting of IEC/TC 25. Decisions from the meeting can be found in doc. N 1196. Regarding proposed joint work, see Decision 2017/25-2:



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Working Group 1 is disbanded. The tasks of WG 1, as related to ISO/IEC 80000 series, are proposed to be transferred to a new Joint Working Group, to be activated together to ISO/TC 12, with the scope “Revision and amendment of IEC-related parts of ISO/IEC 80000 series”. The Secretariat shall inform ISO/TC 12 of this decision and invite it to contribute to the new JWG. In order to effectively deal with its activities in this new scope, the new JWG will be organized into Projects Teams.

Ms. Ferrés explained the modes for cooperation between IEC and ISO.

ISO/TC 12 favoured the proposal to establish a joint working group between ISO/TC 12 and IEC/TC 25. The joint working group will be under IEC lead of IEC/TC 25 because the original standards were developed in IEC/TC 25. It was noted that Mr. Krystek will be the convenor of the Joint Working Group.

Action 3, Berlin 2017: To circulate a call for experts for the new JWG on the revision of IEC 80000-6, *Electromagnetism*, IEC 80000-13, *Information science and technology*

Resolution 106, Berlin 5/2017

9.3 ITU-T/SG 17 – Security

ITU-T/SG 17 did not send a liaison report to this meeting.

9.4 Organizations in liaison

BIPM, CIE, IAEA, ICAO, ICRU, ILO, IUCr, IUPAC, IUPAP, Infoterm, OASIS, OIML, UNECE, WHO, WMO, EC, IMO, WCO

EURAMET

Resolution: Noting the mutual interest from EURAMET and ISO/TC 12 to form liaison. The Liaison Officer from ISO/TC 12 will be Prof. Leslie Pendrill. The Liaison Officer from EURAMET will be Mr. Duncan Jarvis.

Action 4, Berlin 2017: The secretary to see to that the Global Directory is updated regarding the liaison with EURAMET.

Resolution 107, Berlin 6/2017

9.5 Review of list of liaison officers from ISO/TC 12 to liaisons

Doc. ISO/TC 12 N 1154, N 1181

ISO/TC 12 decided to compile the list of committees and organizations in liaison and to circulate it to the members of ISO/TC 12 for comments as to the relevance of the present liaisons.

Action 5, Berlin 2017: The secretary will perform a survey regarding the relevance of liaisons.



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10. Status of all items of the programme of work and action to be taken

Doc. ISO/TC 12 Link to Programme of Work:

<https://www.iso.org/committee/46202/x/catalogue/p/0/u/1/w/0/d/0>

ISO 80000-5, *Thermodynamics*, -9, *Physical chemistry and molecular physics* and -11, *Characteristic numbers* have been completed for FDIS ballot. Part 1, *General*, Part 3: *Space and time* and Part 8, *Acoustics* remain to be completed. WG 19 was urged to continue its work on the resolution of the comments received on Parts 1, 3 and 8.

DIS 80000-1 – *Quantities and units - Part 1: General*, has failed approval in CEN (prEN ISO 80000-1) but was approved in the other organizations. The TPM, Ms. Ferrés, stressed that the technical issues should be resolved.

The French member body was encouraged to change its position on prEN ISO 80000-1 in order not to break the Vienna agreement.

Mr Krystek stated that the document will be in line with the position of CIPM. The issue is the definition of 3.9 "unit of measurement". The change was made in drafts for the SI brochure after December 2013.

Ms. Ferrés proposed that ISO/TC 12 should write to CIPM to ask for an explanation on the change in the SI brochure in the latest edition.

Resolution 108, Berlin 7/2017

Action 6, Berlin 2017: ISO/TC 12 concluded that the chair would write a letter to CIPM asking for a clarification on the change of the revised definition of "unit of measurement".

The chair raised the question of the status of ISO/TC 12 in relation to the other organizations. ISO/TC 12 resolved that the ISO 80000-series should follow the decisions of Metre Convention bodies.

Resolution 109, Berlin 8/2017

The time table for the publication of the ISO 80000 series was discussed. ISO 80000-2, -4, 5, -7, -9, -10, -11, -12 have successfully passed the DIS ballot. Noting the extensive change in the format and presentation of the series, and the interrelations of the various parts, it is necessary to submit all the parts simultaneously to FDIS ballot. Parts -1, -3, and -8 completed the DIS during late 2017, receiving extensive comments. WG 19 plans to complete the resolution of comments on these parts by early 2018. The final drafting and editorial alignment of all the series, to proceed to FDIS ballot, will be achieved by May 2018.

All parts of the ISO 80000-series need to be submitted to ISO/CS **May 1, 2018**.

Resolution 110, Berlin 9/2017

Action 7, Berlin 2017: The secretary was asked to ask TMB for an extension of the deadline for cancellation of all parts of the ISO 80000 series.



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11. Follow-up on work

11.1 ISO/TC 12-N 1168, Proposal to CCU

Doc. ISO/TC 12 N 1177 RoV

Mr Krystek reported that the SI brochure has omitted the units knot and nautical mile. ISO/TC 12 made a survey among its members, doc. N 1168. This documents states:

"The units "nautical mile" and "knot" are no longer included in the new SI brochure (in the old one they are included). This is problematic because these units are used in navigation (marine and aviation, see publications of the International Hydrography Organization (IHO) and the International Civil Aviation Organization (ICAO) as well as ISO 8729-2:2009 and IEC 62238:2003). They also serve to define the territorial boundaries of coastal waters (see United Nations Convention on the Law of the Sea (UNCLOS), http://www.un.org/depts/los/convention_agreements/texts/unclos/part2.htm). Therefore, these units shall be also be included in the new SI brochure."

16 members responded; 6 abstentions, 7 votes for and 3 votes against, see doc. ISO/TC 12 N 1177. This meant that the required majority had been reached. CCU has already been informed of the result.

11.2 ISO-TC 12 N 1165; WD of OIML D5

Doc. ISO/TC 12 N 1176 RoV

The secretary informed that 3 members; Portugal, Russian Federation and Sweden had submitted comments on OIML D 5; *Hierarchy schemes for measuring instruments*. The comments have been edited by the chair and secretary and sent to OIML.

12. Work items on which no progress is being made - Status and action to be taken

All items on the programme of work have progressed since the last meeting of ISO/TC 12.

13. Items for future work

IEC 80000-13, *Information science and technology*, -15, *Logarithmic and related quantities (revision of IEC 60025-3)*, -16, *Printing and writing rules*, and -17, *Time dependency*. Parts 15 and 16 are based on IEC 60027-1, *Letter symbols to be used in electrical technology - Part 1: General*, the titles for parts 15 and 16 are provisional. The documents will be developed in the new Joint Working Group, see § 9.2.

Mr. Krystek remarked that there is a German standard on logarithmic quantities and units, DIN 5493:2013-10.



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14. Requirements concerning a subsequent meeting

Next meeting will be held on October 8, 2018 place to be decided.

WG 19 will meet January 2018.

Resolution 107, Berlin 6/2017

Action 8, Berlin 2017: The secretary to write to the non-European members and ask them if they would consider hosting the next meeting.

15. Any other business

15.1 Meeting feedback survey:

<http://isotc.iso.org/livelink/livelink?func=ll&objId=16501557&objAction=browse&viewType=1>

The secretary informed the delegates that there is a short meeting survey after each ISO/TC meeting. The delegates were encouraged to take the time to fill out the survey in order for the secretariat to make improvements.

15.2 Information on horizontal committees

The TPM, Ms. Ferrés, informed about the brochure for horizontal committees; N 1193.

Action 8, Berlin 2017: The Chairman and secretary will review the text and propose updates if necessary.

16. Approval of resolutions

The resolutions were read out and approved.

17. Closure of the meeting (16:00 h)

ISO/TC 12 thanked the German member, DIN for the hosting of this meeting and the kind hospitality. Special thanks were addressed to Mr. Dominik for the excellent arrangements and kind and effective assistance during the meeting